

Thesis progress report

LLM-Guided Neural Network Compilation for Real-time UAV Edge Inference

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What can be claimed

The path runs from a companion computer that must see, to a measured file on a simulated camera. This report states the claim. It does not add another tool.

The work can say that a language model names an allowlisted plan, that a checker can refuse it, that an engine rewrites ONNX, that a profiler measures, that the kept plan is the fastest one that passed, that one extra try is allowed after measurement, and that ROS only loads the file.

It cannot yet say TensorRT speed, board power, or accuracy on labelled images. Those need a board with an NVIDIA GPU. This machine is QEMU on ARM and has none. Latency from this machine must not be compared with a Jetson.

Every stored run sets `fps_claimed` to false. When a throughput number appears, it is 1000 divided by the mean latency in milliseconds. It is not a camera frame rate, and it is not a claim about a Jetson.

TensorRT was the original target, because that is the usual edge compiler in the UAV papers. This machine is a QEMU ARM guest with a virtual GPU and no NVIDIA device. A TensorRT number here would have been invented. The same discipline applies to power and to accuracy on labelled frames. The claim stops where the evidence stops. The right column of the figure is not a failure of the method. It is the list of experiments that this host cannot run.

Can say now

An allowlisted plan

A real ONNX rewrite

A measured choice on this CPU

One extra try after measurement

ROS only loads the file

Waits for a board

TensorRT, power, labelled accuracy

Figure 1. The left column is the thesis. The right column waits for a board.

TensorRT speed and board power are the measurements that belong on a real board. They are not results of this machine.